

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-274

WASTE DISCHARGE REQUIREMENTS
FOR
CITY AND COUNTY OF SAN FRANCISCO
HETCH HETCHY WATER AND POWER
EARLY INTAKE WASTEWATER SYSTEM
TUOLUMNE COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The City and County of San Francisco owns a wastewater treatment and disposal system that serves the wastewater needs of the Early Intake facility, a hydroelectric generating plant and its related operation and maintenance complex. Hetch Hetchy Water and Power operates the wastewater system.
2. The City and County of San Francisco and Hetch Hetchy Water and Power (hereafter Discharger) submitted a Report of Waste Discharge (RWD), dated 19 July 2001, for the wastewater system at the Early Intake facility. Additional information was submitted on 13 August 2001 and on 25 September 2001.
3. The Early Intake facility consists of Kirkwood Powerhouse (a hydroelectric generating plant) and its related operation and maintenance complex. The facility is in Sections 2 and 11, T1S, R18E, MDB&M, as shown in Attachment A, which is attached hereto and made part of this Order by reference.
4. The Discharger currently discharges its wastewater to an existing septic system, which is not in compliance with Board policy and is not regulated by Waste Discharge Requirements (WDRs). The current wastewater system is near the edge of the Tuolumne River, and is in danger of washing out during a serious flood event. The Discharger has proposed to rehabilitate and modify the existing wastewater system in order to come into compliance with applicable regulations and eliminate the potential of catastrophic failure. The Discharger anticipates that the rehabilitated wastewater system will be completed by September 2002.
5. For the purposes of this Order, the term "wastewater system" shall mean the domestic sewage and industrial wastewater collection and transport system, the oil/water separator, septic tanks, and the leachfield disposal area.
6. The layout of the facility (including the basic components of the existing and proposed wastewater system) is shown on Attachment B, which is attached hereto and made part of this Order by reference.

Sources of Wastewater

7. The wastewater system receives its domestic waste from various residential and facility-related sources. These include seven cottages, a bunkhouse, a dining hall, offices, a pool bathroom, and the switchyard bathroom. The projected average daily flow of domestic waste into the system is estimated to be approximately 1,500 gallons per day (gpd).
8. The Kirkwood Powerhouse generates approximately 1,500 gpd of industrial wastewater consisting of water that seeps into the powerhouse. This water is collected by floor drains throughout the powerhouse and accumulates in a drainage sump located in the bottom floor of the powerhouse. It contains a small quantity of food grade grease and oil and is passed through an oil/water separator. According to the Discharger, no detectable levels of hydrocarbon-related constituents have been found in the discharge from the separator. The capacity of the oil/water separator is 8,000 gallons per minute.
9. The Discharger projects that the average dry weather flow into the entire wastewater system will be approximately 3,000 gpd. The projected peak flow into the system is 5,000 gpd.

Proposed System

10. The proposed modified wastewater system will consist of a 10,000 gallon septic tank, a new wastewater collection and transport system, and a mounded intermittent sand filter leachfield. The design capacity of the system is 5,000 gpd.
11. The proposed leachfield will be west of the switchyard on a bench that is approximately 75 feet above the river, with a horizontal distance of over 100 feet from the river. The bench area is approximately 250 feet long and 60 feet wide and consists of well-graded gravels, with some fines. The average percolation rate of the gravel was determined to be approximately 2 minutes per inch. Based on a 1997 soils investigation report, groundwater was encountered at a depth of approximately 75 feet below the ground surface of the bench. The underlying bedrock is granitic.
12. Because of the fast percolation rate, the Discharger has proposed an engineered intermittent sand filter leachfield system. The sand filter will be approximately 200 feet long by 40 feet wide, and will consist of a 3 foot thick bed of sand, overlain by a 1.5 foot thick layer of gravel. The leachfield perimeter will be lined with a 40 ml HDPE lining material. Leachfield laterals will be placed in the gravel layer, and a pressurized dosing system will be utilized in order to ensure even wastewater distribution and to extend the life of the leachfield.
13. Eighteen observation wells will be systematically spaced throughout the leachfield in order to monitor wastewater levels within the leachfield. The level of water in the leachfield observation wells will be monitored and recorded weekly. Observations made during these monitorings will be recorded and submitted to the Board as part of the quarterly monitoring reports. If the water level in any of the wells is found to be within three feet of the natural ground surface, daily monitoring

will be conducted. If the water level rises to within two feet of the natural ground surface, discharge to the leachfield shall cease and zero discharge maintained. The Discharger must report it to the Board within twenty-four hours if any of the above-described high water conditions should develop.

14. The Discharger proposes to use a commercial septage hauler to pump out the septic tank and haul the waste to facility duly authorized to dispose of septage.

Sanitary Sewer System

15. The rehabilitated wastewater system will collect and transport wastewater through the use of sewers, pipes, pumps, lift stations, and/or other conveyance systems and ultimately direct wastewater to the leachfield. The sewer line will cross the Tuolumne River to deliver the wastewater to a pump station and ultimately to the leachlines. The sewer line which crosses the river will be a double containment line which will be securely attached to the decking and framework of the vehicular bridge which crosses the river, in order to prevent breakage during a major flood event.
16. A "sanitary sewer overflow" occurs when there is a discharge to ground or surface water from the wastewater collection and transport system at any point upstream of the leachfield. Temporary storage and conveyance facilities (such as wet wells, lift stations, tanks, etc.) may be a part of the sanitary sewer system and discharges to these facilities are not considered sanitary sewer overflows, provided that the waste is fully contained within these temporary storage/conveyance facilities.
17. Sanitary sewer overflows consist of varying mixtures of domestic sewage and industrial wastewater; this mixture depends on pattern of use in the sewage collection system tributary to the overflow. The chief causes of sanitary sewer overflows include grease blockages, root blockages, debris blockages, sewer line flood damage, manhole structure failures, vandalism, pump station mechanical failures, power outages, storm or groundwater inflow/infiltration, lack of capacity, and contractor caused blockages.
18. Sanitary sewer overflows often contain high levels of suspended solids, pathogenic organisms, toxic pollutants, nutrients, oxygen demanding organic compounds, oil and grease, and other pollutants. Sanitary sewer overflows can cause temporary exceedances of applicable water quality objectives, pose a threat to public health, adversely affect aquatic life, and impair the public recreational use and aesthetic enjoyment of surface waters in the area.
19. The Discharger is expected to take all necessary steps to adequately maintain, operate, and prevent discharges from its sanitary sewer collection system. This Order requires the Discharger to prepare and implement a *Sanitary Sewer System Operation, Maintenance, Overflow Prevention, and Response Plan*.

Regulatory Considerations

20. Federal regulations for stormwater discharges were promulgated by the U.S. Environmental Protection Agency on 16 November 1990 (40 CFR Parts 122, 123, and 124). The regulations require specific categories of facilities which discharge stormwater associated with industrial activities to obtain NPDES permits. The Discharger has developed and implemented a Storm Water Pollution Prevention Plan.
21. National Forest lands surround the Early Intake facility. The lands are used for many purposes, including recreation.
22. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for waters of the Basins. These requirements implement the Basin Plan.
23. Surface water drainage is to the main fork of the Tuolumne River.
24. The beneficial uses of the Tuolumne River are municipal and domestic supply, agricultural supply, power generation; contact and non-contact recreation; freshwater habitat; and wildlife habitat.
25. Specific information regarding groundwater quality in the vicinity of the treatment plant is not available. The beneficial uses of underlying groundwater are municipal, industrial, and agricultural supply.
26. The Board has considered antidegradation pursuant to State Board Resolution No. 68-16 and finds that the discharge is consistent with those provisions, and will not cause an increase in groundwater constituents above that of background levels.
27. On 2 August 1999, the Planning Department of the City and County of San Francisco issued a *Certificate of Determination of Exemption/Exclusion from Environmental Review* for this project. The determination was made that the proposed improvements to this facility are exempt from the provisions of the California Environmental Quality (CEQA), in accordance with Title 14, California Code of Regulations (CCR), Section 15301(b,d).
28. Pursuant to California Water Code (CWC) Section 13263(g), discharge is a privilege, not a right, and adoption of this Order does not create a vested right to continue the discharge.
29. This discharge of wastewater is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 2005, et seq., (hereafter Title 27). The exemption pursuant to Section 20090(b), is based on the following:
 - a. The Board is issuing waste discharge requirements,

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- b. The discharge complies with the Basin Plan, and
 - c. The wastewater does not need to be managed according to Title 22 CCR, Division 4.5, and Chapter 11, as a hazardous waste.
30. Section 13267(b) of CWC provides that: "In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could affect the quality of the waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the board requires. The burden, including costs of these reports, shall bear a reasonable relationship to the need for the reports and the benefits to be obtained from the reports."
30. The California Department of Water Resources set standards for the construction and destruction of groundwater wells (hereafter DWR Well Standards), as described in *California Well Standards Bulletin 74-90* (June 1991) and *Water Well Standards: State of California Bulletin 94-81* (December 1981). These standards, and any more stringent standards adopted by the Discharger or county pursuant to CWC Section 13801, apply to all monitoring wells.

Public Notice

31. The Board consulted with the Tuolumne County Environmental Health Department, and elicited their recommendations regarding public health aspects of the on-site disposal of wastewater.
32. The Board has notified the Discharger, and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
33. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that the City and County of San Francisco and Hetch Hetchy Water and Power, their agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

(Note: Other prohibitions, conditions, definitions, and the method of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated 1 March 1991 that are incorporated by reference into this Order.)

A. Discharge Prohibitions

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. The discharge of wastewater from the sanitary sewer system at any point upstream of the leachfield is prohibited.
4. The surfacing of wastewater outside or downgradient of the leachfield is prohibited.
5. The discharge of waste classified as 'hazardous,' as defined in Section 20164 of Title 27, CCR, or 'designated,' as defined in Section 13173 of the California Water Code, is prohibited.
6. Discharge to the leachfield is prohibited if groundwater rises to within two feet of the ground surface.

B. Discharge Specifications

1. The monthly average daily inflow into the leachfield shall not exceed 3,000 gpd.
2. The maximum daily inflow into the leachfield shall not exceed 5,000 gpd.
3. Discharge from the oil/water separator shall result in no detectable concentrations of total petroleum hydrocarbons (TPH) - gasoline, TPH-diesel, oil & grease, or volatile organic compounds to the wastewater collection system.
4. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater collection, treatment and disposal areas.
5. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.
6. The leachfield shall have enough capacity to accommodate allowable wastewater flow as well as inflow and infiltration during the wet season.
7. The discharge to the leachfield shall cease when the level of water in any of the monitored observation wells in the leachfield rises to within two feet of ground surface.
8. The entire wastewater system, excluding the wastewater pipe which crosses the Tuolumne River, shall be protected to prevent inundation or washout due to floods with a 100-year return frequency.

C. Solids Disposal Requirements

Sludge, as used in this document, means the solid, semisolid, and liquid residues removed during the wastewater treatment processes. Solid waste refers to grit and screening material generated during preliminary treatment. Residual sludge means sludge that will not be subject to further treatment at this facility.

1. Sludge and solid waste shall be removed from septic tanks, screens, clarifiers, etc. as needed to ensure optimal plant operation.
2. Any storage of residual sludge and solid waste on property of the facility shall be temporary and controlled and contained in a manner that minimizes leachate formation and precludes infiltration of waste constituents into soils in a mass or concentration that will violate Groundwater Limitations.
3. Residual sludge and solid waste shall be disposed of in a manner approved by the Executive Officer and consistent with Title 27. Removal for further treatment, disposal, or reuse at sites (i.e., landfill, WWTF, composting site) operated in accordance with valid waste discharge requirements issued by a regional water quality control board will satisfy this specification.

D. Groundwater Limitations

The discharge, in combination with other site derived sources, shall not cause underlying groundwater to contain waste constituents in concentration statistically greater than background water quality, except for coliform bacteria. For coliform bacteria, increases shall not cause the most probable number of total coliform organisms to exceed 2.2 MPN/100 ml over any 7-day period.

E. Provisions

1. In accordance with California Business and Professions Code sections 6735, 7835, and 7835.1, engineering and geologic evaluations and judgments shall be performed by or under the direction of registered professionals competent and proficient in the fields pertinent to the required activities. All technical reports specified herein that contain work plans for, that describe the conduct of investigations and studies, or that contain technical conclusions and recommendations concerning engineering and geology shall be prepared by or under the direction of appropriately qualified professional(s), even if not explicitly stated. Each technical report submitted by the Discharger shall contain a statement of qualifications of the responsible licensed professional(s) as well as the professional's signature and/or stamp of the seal.
2. All of the following reports shall be submitted pursuant to Section 13267 of the California Water Code and shall be prepared as described by Provision 1.

- a. By **1 March 2002**, the Discharger shall submit a workplan for characterization of groundwater quality. The workplan shall describe the installation of wells to allow evaluation of the groundwater quality upgradient and downgradient of the leachfield area. Every monitoring well shall be constructed to yield representative samples from the uppermost layer of the uppermost aquifer and to comply with applicable well standards. The workplan shall be consistent with, and include the items listed in, the first section of Attachment C, *"Items to be Included in a Monitoring Well Installation Workplan and a Monitoring Well Installation Report of Results."*
- b. By **1 May 2002**, the Discharger shall submit a *Sanitary Sewer System Operation, Maintenance, Overflow Prevention, and Response Plan* (SSS Plan) that describes the actions designed to prevent, or minimize the potential for sanitary sewer overflows. The Discharger shall maintain the SSS Plan in an up-to-date condition and shall amend the SSS Plan whenever there is a change (e.g. in the design, construction, operation, or maintenance of the sanitary sewer system or sewer facilities) that materially affects the potential for sanitary sewer overflows, or whenever there is a sanitary sewer overflow. The Discharger shall ensure that the up-to-date SSS Plan is readily available to sewer system personnel at all times and that sewer system personnel are familiar with it.
 - i. At a minimum, the Collection System Operation and Maintenance portion of the plan shall contain or describe the following:
 1. Detailed maps of the sanitary sewer system, identifying sewer mains, manholes, and lift stations;
 2. A detailed listing of elements to be inspected, a description of inspection procedures and inspection frequency, and sample inspection forms;
 3. Routine inspection and testing of all pipelines, lift stations, valves, and other key system components. The inspection/testing program shall be designed to reveal problems that might lead to accidental spills and ensure that preventive maintenance is completed;
 4. Repair or replacement of old, worn out, or defective equipment;
 5. Minimize the need for manual operation of critical systems and provide spill alarms or other "fail safe" mechanisms;
 6. Properly manage, operate and maintain, at all times, all parts of the collection system that the Discharger owns or over which the Discharger has operational control;

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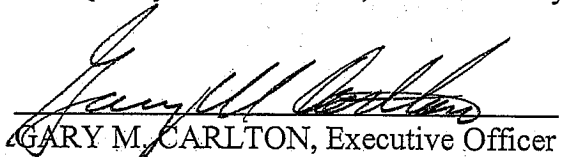
7. Provide adequate capacity to convey base flows and peak flows for all parts of the collection system the Discharger owns or over which the Discharger has operational control; and
 8. Take all feasible steps to stop and mitigate the impact of sanitary sewer overflows in portions of the collection system the Discharger owns or over which the Discharger has operational control; and
- ii. At a minimum, the Overflow Prevention and Response Plan shall contain or describe the following:
1. Identification of areas of the collection system that have a potential to overflow;
 2. Maintenance activities that can be implemented to address the cause of the overflow and means to prevent overflows. Maintenance activities may include pretreatment of wastewater from industrial dischargers who discharge high concentrations of oil and grease in their wastewater;
 3. Procedures for responding to sanitary sewer overflows designed to minimize the volume of sewer overflow that enters surface waters, and minimize the adverse effects of sewer overflows on water quality and beneficial uses;
 4. Describe steps to be taken when an overflow or spill occurs, and procedures that will be implemented to ensure that all overflows and spills are properly identified, responded to and reported; and
 5. A public notification plan, in which any posting of areas contaminated with sewage is performed at the direction of the Tuolumne County Health Department. All parties with a reasonable potential for exposure to an overflow event shall be notified.
- c. **Thirty days prior to beginning operation of the new system**, the Discharger shall submit a groundwater well installation report. The report shall be consistent with the guidance in the second section of Attachment C.
- d. **Thirty days after beginning operation of the new system**, the Discharger shall submit a report certifying that the system was constructed as described in these WDRs. The report shall include as-built drawings of the system and shall propose sample locations within the leachfield.
3. The Discharger shall comply with the *Standard Provisions and Reporting Requirements for Waste Discharge Requirements*, dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as *Standard Provision(s)*.

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4. The Discharger shall comply with Monitoring and Reporting Program No. 5-01-274, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
5. At least **90 days prior** to termination or expiration of any lease, contract, or agreement involving the wastewater collection, treatment, or disposal system, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
6. The Discharger shall use best practicable treatment or control, including proper operation and maintenance, to comply with terms of this Order.
7. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
8. The Discharger must comply with all applicable conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
9. A copy of this Order, including its attachments and Standard Provisions, shall be kept at the wastewater treatment facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
10. The Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 7 December 2001.


GARY M. CARLTON, Executive Officer

JRM:12/7/01

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This Monitoring and Reporting Program (MRP) describes requirements for monitoring the oil/water separator effluent, septic tank effluent, the leachfield effluent, and groundwater. This MRP is issued pursuant to Water Code Section 13267. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Executive Officer. Regional Board staff shall approve specific sample station locations prior to implementation of sampling activities.

This MRP is effective upon date of signature; however, samples need not be collected until the rehabilitated wastewater system has been constructed and is in use. In the meantime, the Discharger shall submit quarterly progress reports, as described in the "Reporting" section of this MRP.

All samples should be representative of the volume and nature of the discharge or matrix of material sampled. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form.

OIL/WATER SEPARATOR EFFLUENT MONITORING

Effluent samples shall be collected at the outlet from the oil/water separator. Effluent monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Daily Flow	Gpd	Meter	Continuously	Quarterly
Monthly Average Daily Flow	Gpd	Calculated	Monthly	Quarterly
Oil & Grease (Method 5520C/1664)	mg/l	Grab	Monthly	Quarterly
TPH-G (EPA Method 8015/5030)	mg/l	Grab	Monthly	Quarterly
TPH-D (EPA Method 8015/3510)	mg/l	Grab	Monthly	Quarterly
VOCs (EPA Method 8021B)	ug/l	Grab	Monthly	Quarterly

SEPTIC TANK EFFLUENT MONITORING

Septic tank effluent samples shall be collected that are representative of the effluent for the period sampled. Septic tank effluent monitoring shall include the following:

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<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Daily Flow	gpd	Meter	Continuously	Quarterly
Monthly Average Daily Flow	gpd	Calculated	Monthly	Quarterly
BOD ₅ ¹	Mg/l	Grab	Quarterly	Quarterly
Total Suspended Solids	Mg/l	Grab	Quarterly	Quarterly

¹ Five-day, 20° Celsius biochemical oxygen demand

LEACHFIELD EFFLUENT MONITORING

Effluent samples shall be collected from the leachfield observation wells and shall be representative of the volume and nature of the discharge. All observation wells shall be sampled for depth to upper water surface, while only two observation wells shall be sampled for the remaining constituents. Regional Board staff shall approve specific sample station locations prior to implementation of sampling activities. Depth to upper water surface shall be measured to the nearest 0.1 feet. Leachfield effluent monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Depth to Upper Water Surface	0.1 feet	Measurement	Weekly ¹	Quarterly
BOD ²	mg/l	Grab	Quarterly	Quarterly
Total Coliform Organisms ³	MPN ⁴ /100 ml	Grab	Quarterly	Quarterly
Total Dissolved Solids	mg/l	Grab	Quarterly	Quarterly
Nitrate-Nitrogen	mg/l	Grab	Quarterly	Quarterly
Total Kjeldahl Nitrogen	mg/l	Grab	Quarterly	Quarterly

¹ If the water level in any of the wells is found to be within three feet of ground surface, daily monitoring shall be implemented, and the condition shall be reported to the Board within 24 hours.

² 5-day Biochemical Oxygen Demand

³ Method No. 9221E, using a minimum of five dilutions or 25 tubes

⁴ Most Probable Number

GROUNDWATER MONITORING

Prior to construction and/or sampling of any groundwater monitoring wells, the Discharger shall submit plans and specifications to the Board for review and approval. Once installed, all new wells shall be added to the MRP and shall be sampled and analyzed according to the schedule below.

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Prior to sampling, the groundwater elevations shall be measured and the wells shall be purged at least three well volumes until temperature, pH and electrical conductivity have stabilized. Depth to groundwater shall be measured to the nearest 0.01 feet. Samples shall be collected using standard EPA methods. Groundwater monitoring shall include, at a minimum, the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
Depth to Groundwater	0.01 feet	Measurement	Quarterly
Groundwater Elevation	0.01 feet	Calculated	Quarterly
Gradient	feet/feet	Calculated	Quarterly
Gradient Direction	degrees	Calculated	Quarterly
Total Dissolved Solids	mg/l	Grab	Quarterly
Nitrate-nitrogen	mg/l	Grab	Quarterly
Total Coliform Organisms ¹	MPN/100 ml	Grab	Quarterly
Total Kjeldahl Nitrogen	mg/l	Grab	Quarterly
Oil & Grease (Method 5520C/1664)	mg/l	Grab	Quarterly ²
TPH-G (EPA Method 8015/5030)	mg/l	Grab	Quarterly ²
TPH-D (EPA Method 8015/3510)	mg/l	Grab	Quarterly ²
VOCs (EPA Method 8021B)	Ug/l	Grab	Quarterly ²

¹ Method No. 9221E, using a minimum of 15 tubes or three dilutions.

² To be sampled for these constituents only if Oil & Grease, TPH-G, TPH-D, or VOCs were detected in the oil/water separator effluent during the previous quarter's monitoring

FACILITY MONITORING

All portions of the wastewater system, including septic tanks, manholes, and lift stations shall be inspected on a quarterly basis. The ground surface above and around the leachfield shall be inspected on a twice monthly basis. Observations made during these inspections shall be recorded and submitted to the Board as part of the quarterly monitoring report.

SLUDGE MONITORING

A log shall be kept of sludge quantities removed and of handling and disposal activities. At a minimum, the depth of sludge in the septic tank shall be measured and reported to the Board on a quarterly basis.

REPORTING

In reporting monitoring data, the District shall arrange the data in tabular form so that the date, sample type (e.g., effluent, groundwater, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Regional Board.

As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all Groundwater Monitoring Reports shall be prepared under the direct supervision of a Registered Engineer or Geologist and signed by the registered professional.

A. Quarterly Monitoring Reports

Daily, weekly, monthly, and quarterly monitoring data shall be reported in quarterly monitoring reports. Quarterly reports shall be submitted to the Regional Board on the **1st day of the second month after the quarter** (i.e. the January – March quarter report is due by 1 May). At a minimum, the reports shall include:

1. If the rehabilitated wastewater system is not yet operational, then the report shall describe the construction progress to date and the anticipated date of operation.
2. If the rehabilitated wastewater system has been constructed and is operational, the report shall contain:
 - a.) Results of facility, sludge, oil/water separator effluent, septic tank effluent, leachfield effluent, and groundwater monitoring;
 - b.) A comparison of monitoring data to the discharge specifications and/or water quality goals, and an explanation of any violations. Data shall be presented in tabular format;
 - c.) If requested by staff, copies of laboratory analytical report(s);
 - d.) A narrative description of all preparatory, monitoring, sampling, and analytical testing activities for the groundwater monitoring. The narrative shall be sufficiently detailed to verify compliance with the WDR, this MRP, and the Standard Provisions and Reporting Requirements. The narrative shall be supported by field logs for each well documenting depth to groundwater; parameters measured before, during, and after purging; method of purging; calculation of casing volume; and total volume of water purged;

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- e.) Calculation of groundwater elevations, an assessment of groundwater flow direction and gradient on the date of measurement, comparison of previous flow direction and gradient data, and discussion of seasonal trends if any;
- f.) A narrative discussion of the analytical results for all groundwater locations monitored including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable);
- g.) A comparison of monitoring data to the oil/water separator effluent and groundwater limitations, and an explanation of any violation;
- h.) Summary data tables of historical and current water table elevations and analytical results;
- i.) A scaled map showing roads, leachfield location, observation well locations, the locations of monitoring wells, and groundwater elevation contours referenced to mean sea level datum;
- j.) Copies of laboratory analytical report(s) for groundwater monitoring.

A letter transmitting the self-monitoring reports shall accompany each report. Such a letter shall include a discussion of requirement violations found during the reporting period, and actions taken or planned for correcting noted violations, such as operation or facility modifications. If the discharger has previously submitted a report describing corrective actions and/or a time schedule for implementing the corrective actions, reference to the previous correspondence will be satisfactory. The transmittal letter shall contain a statement by the discharger, or the discharger's authorized agent, under penalty of perjury, that to the best of the signer's knowledge the report is true, accurate and complete.

The Discharger shall implement the above monitoring program as of this date.

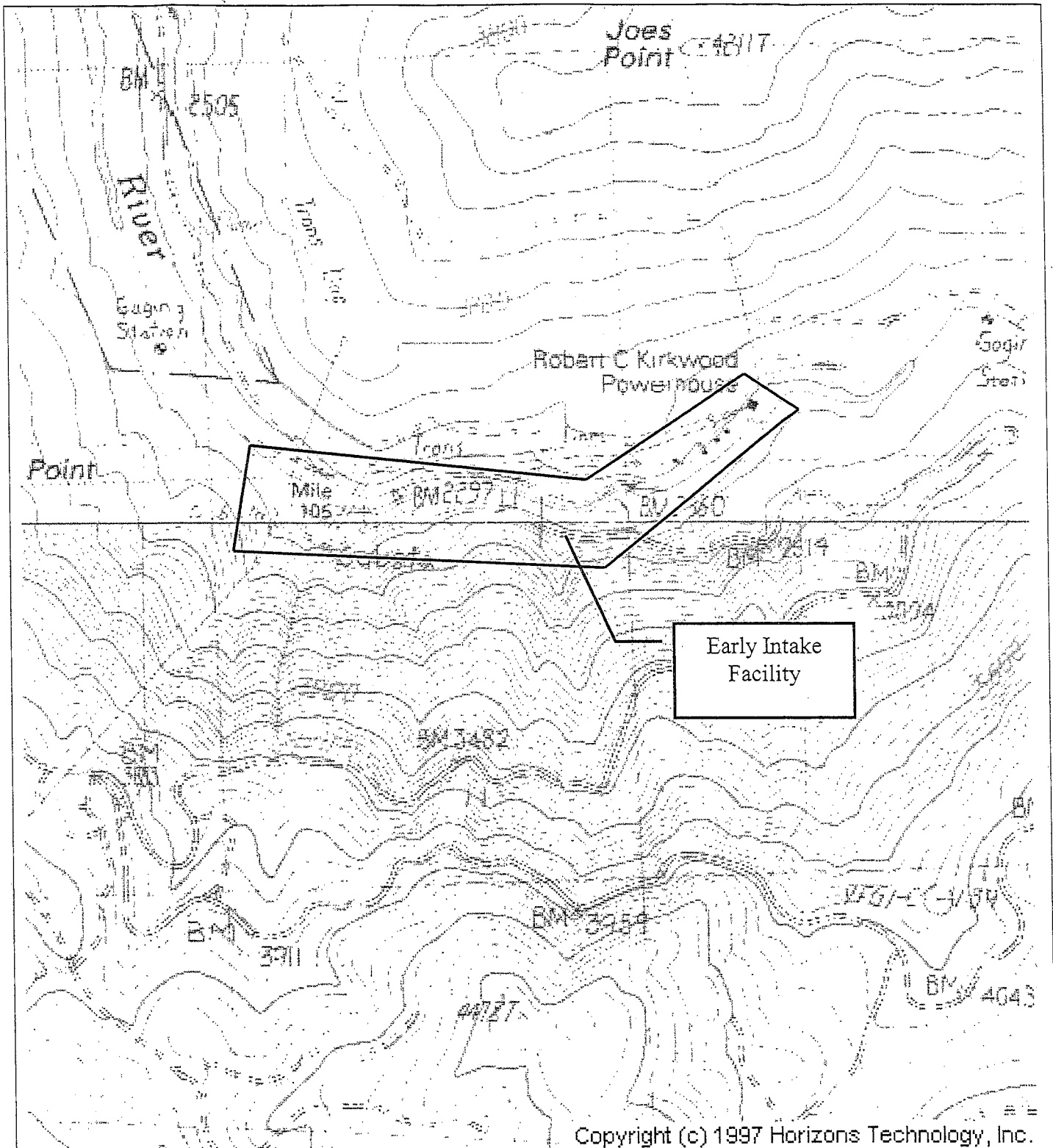
Ordered by:


GARY M. CARLTON, Executive Officer

7 December 2001

(Date)

JRM:12/7/01

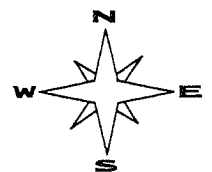


Drawing Reference:

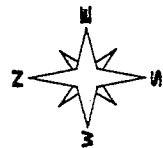
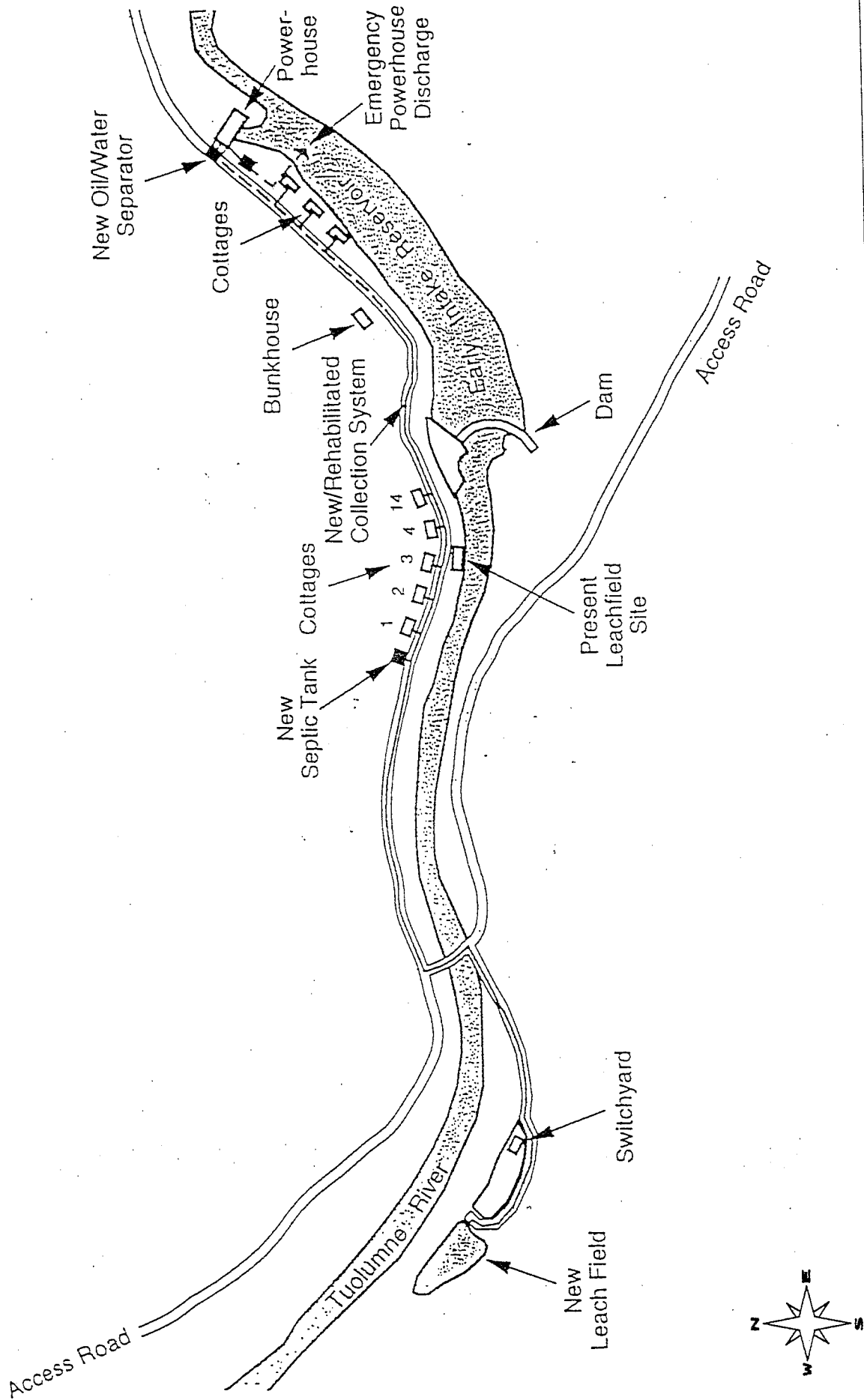
HETCH HETCHY
QUADRANGLE
U.S.G.S TOPOGRAPHIC MAP
7.5 MINUTE QUADRANGLE

SITE LOCATION MAP

CITY AND COUNTY OF SAN FRANCISCO
EARLY INTAKE WASTEWATER SYSTEM
WASTE DISCHARGE REQUIREMENTS
ORDER NO. 5-01-274



approx. scale
1 in. = 1,000 ft.



SITE PLAN

CITY AND COUNTY OF SAN FRANCISCO
 EARLY INTAKE WASTEWATER SYSTEM
 WASTE DISCHARGE REQUIREMENTS
 ORDER NO. 5-01-274

Drawing Reference:
 Camp Dresser & McKee
 Early Intake Vicinity Map
 February 1995

Approximate Scale
 1 in = 500 feet

ORDER NO. 5-01-274
ATTACHMENT C

**ITEMS TO BE INCLUDED IN A
MONITORING WELL INSTALLATION WORKPLAN AND A
MONITORING WELL INSTALLATION REPORT OF RESULTS**

Prior to installation of groundwater monitoring wells, the Discharger shall submit a workplan containing the minimum listed information. Wells may be installed after staff approve the workplan. Upon installation of the monitoring wells, the Discharger shall submit a report of results, as described below. All workplans and reports must be signed by a registered geologist, certified engineering geologist, or civil engineer registered or certified by the State of California.

Monitoring Well Installation Workplan

A. General Information:

- Monitoring well locations and rationale
- Survey details
- Equipment decontamination procedures
- Health and safety plan
- Topographic map showing any existing monitoring wells, proposed wells, waste handling facilities, utilities, and other major physical and man-made features.

B. Drilling Details: describe drilling and logging methods

C. Monitoring Well Design:

- Casing diameter
- Borehole diameter
- Depth of surface seal
- Well construction materials
- Diagram of well construction
- Type of well cap
- Size of perforations and rationale
- Grain size of sand pack and rationale
- Thickness and position of bentonite seal and sand pack
- Depth of well, length and position of perforated interval

D. Well Development:

- Method of development to be used
- Method of determining when development is complete
- Method of development water disposal

E. Surveying Details: discuss how each well will be surveyed to a common reference point

F. Soil Sampling (if applicable):

- Cuttings disposal method
- Analyses to be run and methods
- Sample collection and preservation method
- Intervals at which soil samples are to be collected
- Number of soil samples to be analyzed and rationale
- Location of soil samples and rationale
- QA/QC procedures

G. Well Sampling:

- Minimum time after development before sampling (48 hours)
- Well purging method and amount of purge water
- Sample collection and preservation method
- QA/QC procedures

H. Water Level Measurement:

- The elevation reference point at each monitoring well shall be within 0.01 foot.
- Ground surface elevation at each monitoring well shall be within 0.1 foot.
- Method and time of water level measurement shall be specified.

I. Proposed time schedule for work.

Monitoring Well Installation Report of Results

A. Well Construction:

- Number and depth of wells drilled
- Date(s) wells drilled
- Description of drilling and construction
- Approximate locations relative to facility site(s)
- A well construction diagram for each well must be included in the report, and should contain

the following details:

- Total depth drilled
- Depth of open hole (same as total depth drilled if no caving occurs)
- Footage of hole collapsed
- Length of slotted casing installed
- Depth of bottom of casing
- Depth to top of sand pack
- Thickness of sand pack
- Depth to top of bentonite seal
- Thickness of bentonite seal
- Thickness of concrete grout
- Boring diameter
- Casing diameter
- Casing material

Size of perforations
Number of bags of sand
Well elevation at top of casing
Depth to ground water
Date of water level measurement
Monitoring well number
Date drilled
Location

B. Well Development:

Date(s) of development of each well
Method of development
Volume of water purged from well
How well development completion was determined
Method of effluent disposal
Field notes from well development should be included in report.

C. Well Surveying: provide reference elevations for each well and surveyor's notes

D. Water Sampling:

Date(s) of sampling
How well was purged
How many well volumes purged
Levels of temperature, EC, and pH at stabilization
Sample collection, handling, and preservation methods
Sample identification
Analytical methods used
Laboratory analytical data sheets
Water level elevation(s)
Groundwater contour map

E. Soil Sampling (if applicable):

Date(s) of sampling
Sample collection, handling, and preservation method
Sample identification
Analytical methods used
Laboratory analytical data sheets

INFORMATION SHEET

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-01-274
CITY AND COUNTY OF SAN FRANCISCO
HETCH HETCHY WATER AND POWER
EARLY INTAKE WASTEWATER SYSTEM
TUOLUMNE COUNTY

The City and County of San Francisco (hereafter Discharger) operate a hydroelectric generation plant (Kirkwood Powerhouse) and its related operations and management complex on the main fork of the Tuolumne River. The facility generates domestic and industrial wastewater. Domestic wastewater originates from seven cottages, a bunkhouse, a kitchen and dining hall, offices, a pool bathroom, and a switchyard bathroom. The facility generates approximately 1,500 gallons per day (gpd) of industrial wastewater consisting of water that seeps into the powerhouse. This water is collected by floor drains throughout the powerhouse and accumulates in a drainage sump located in the bottom floor of the powerhouse. It may contain a small quantity of food grade grease and oil and is passed through an oil/water separator. According to the Discharger, no detectable levels of hydrocarbon-related constituents have been found in the discharge from the separator. The average dry weather flows are estimated to be approximately 3,000 gallons per day (gpd), with peak flows estimated to be approximately 5,000 gpd.

The Discharger currently discharges its wastewater to an existing septic system, which is not in compliance with Board policy and is not regulated by Waste Discharge Requirements (WDRs). The current wastewater system is near the edge of the Tuolumne River, and is in danger of washing out during a serious flood event. The Discharger has proposed to rehabilitate and modify the existing wastewater system in order to come into compliance with applicable regulations and eliminate the potential of catastrophic failure.

The proposed rehabilitated wastewater system will consist of a rehabilitated wastewater collection and transport system, a 10,000 gallon septic tank, and a mounded intermittent sand filter leachfield. The design capacity of the system is 5,000 gpd.

The WDRs specify a monthly average dry weather flow limit of 3,000 gpd and a maximum flow limit of 5,000 gpd. If the water level in the leachfield rises to within two feet of the ground surface, discharge to the leachfield must cease. The WDRs also require the submittal and implementation of a *Sanitary Sewer System Operation, Maintenance, Overflow Prevention and Response Plan*.

The Monitoring and Reporting Program includes requirements for monitoring the oil/water separator discharge, septic tank and leachfield effluent, and groundwater. In order to determine compliance with the groundwater limitation contained herein, these WDRs contain a time schedule for development and implementation of a groundwater monitoring program.

Surface water drainage is to the Tuolumne River.

